



Prospective Associations Between Early Adolescent Problematic Screen Use, Mental Health, Sleep, and Substance Use

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Introduction: There are limited large-scale, prospective analyses examining problematic (addiction-like) screen use and mental and behavioral health outcomes in early adolescents. This study aimed to determine the associations between problematic screen use and mental and behavioral health outcomes—such as depressive symptoms, suicidal behaviors, sleep disturbance, and substance use initiation—1 year later in a national cohort of children aged 11–12 years in the U.S.

Methods: Prospective cohort data from Adolescent Brain Cognitive Development Study participants who reported their screen use at ages 11–12 (Year 2, 2018–2020; N=8,119; 47.5% female; 43.8% racial/ethnic minority) were analyzed in 2025. Associations between problematic screen use (Social Media Addiction Questionnaire, Video Game Addiction Questionnaire, Mobile Phone Involvement Questionnaire, Year 2) and mental and behavioral health outcomes (Year 3) were determined using either linear or logistic regression analyses on the basis of the outcome, adjusting for potential confounders (Year 2).

Results: Problematic mobile phone use and social media use were prospectively associated with higher depressive, somatic, attention/deficit, oppositional defiant, and conduct problems scores; suicidal behaviors; sleep disturbance; and substance initiation. Problematic video game use was associated with higher depressive, attention/deficit, and oppositional defiant scores; suicidal behaviors; and sleep disturbance.

Conclusions: Early adolescent problematic screen use is prospectively associated with poor mental and behavioral health outcomes 1 year later, extending prior cross-sectional findings. Clinicians can assess not only screen time but also problematic screen use and provide guidance to adolescents and parents, including the development of a family media plan.

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INTRODUCTION

Screen time increases in early adolescence,¹ with even more substantial increases apparent during the coronavirus disease 2019 (COVID-19) pandemic.² Screen time encompasses various modalities, including mobile phones, social media, and video games; with an increase in screen time comes an increased risk of problematic screen use (PSU), characterized by addiction-like symptoms such as preoccupation with and compulsion to use, loss of control, and interference with

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0749-3797/\$36.00

<https://doi.org/10.1016/j.amepre.2025.108248>

functioning.^{3,4} More specifically, problematic social media use and mobile phone use are characterized by the perceived need to be online and connected to technology constantly, with mobile phone use also involving activities such as texting, emailing, and video chatting.^{4–6} Problematic video game use is characterized by loss of control over gaming and interference with functioning.^{3,7}

Globally, mental health disorders often onset in adolescence, with one third of mental disorders developing by age 14 years and half by age 18 years.⁸ An estimated 49.5% of U.S. adolescents have experienced mental illness.⁹ Poor sleep is closely related to mental health, representing both a risk factor for the development of mental health disorders¹⁰ and a symptom of such disorders.¹¹ Substance use also often co-occurs with mental health disorders.¹² National surveys have found a lifetime prevalence of substance use among adolescents to be 26.3% for alcohol, 15.4% for cannabis, and 13.4% for tobacco.¹³

The U.S. Surgeon General issued a 2023 Advisory on Social Media and Youth Mental Health, which identified critical evidence gaps, including the need for (1) prospective study designs because most prior research has been cross-sectional; (2) a focus on adolescents because most prior research has focused on young adults or adults; and (3) investigations of problematic use, which could be related to addiction.¹⁴ This study aims to directly address these 3 evidence gaps, by prospectively studying PSU in U.S. early adolescents.

A growing body of work that is mostly cross-sectional has found associations between PSU across different modalities and a range of poor mental and behavioral health outcomes in children. For example, problematic social media use has been associated with poor mental health,^{4,15–17} sleep disturbance,¹⁵ and substance use.¹⁸ Similarly, problematic mobile phone use^{5,19–22} and problematic video game use^{3,16,23–27} have been associated with adverse mental health and poor sleep outcomes. Findings are mixed with regard to associations with substance use.^{5,28–31}

Previous literature has focused primarily on older adolescents (i.e., those aged 15–19 years) and adults. However, early adolescence represents an important time period when individuals are expanding their screen usage and often have their first experiences with mental health concerns and substance use initiation. Furthermore, studies have featured mostly cross-sectional analyses,^{5,16,18,32–37} examined populations outside the U.S.,^{5,16,18,19,24,25,32,33,36,37} and included smaller study sizes than this study.^{18,33,35,24,32,38} This study examined prospective associations between PSU across 3 domains (mobile phones, social media, and video gaming) and mental health, sleep, and substance use outcomes in a

large, demographically diverse cohort of early adolescents aged 11–12 years in the U.S.

METHODS

Study Population

A prospective analysis of data from Year 2 and Year 3 follow-up of the Adolescent Brain Cognitive Development (ABCD) Study (5.0 release) was conducted. The ABCD Study is the largest longitudinal study of adolescent health, brain, and cognitive development in the U.S. that recruited 11,875 children from 21 sites in 2016–2018 (Barch et al.,³⁹ Dick and colleagues,⁴⁰ and Garavan et al.⁴¹ provide details). Participants were mostly aged 11–12 years during the Year 2 follow-up (2018–2020). The analytic sample in this study included 8,119 participants who had complete sociodemographic data; reported any mobile phone, social media, or video game use; and provided data for at least 1 outcome. The excluded participants were more likely to be younger, to be female, to have high income, and to have high parental education and less likely to be White ([Appendix Table 1](#), available online). Centralized IRB approval was received from the University of California, San Diego. Written assent and consent were obtained from study participants and parents/guardians, respectively.

Measures

Regarding PSU exposures ([Appendix Table 2](#), available online, provides questions), the self-report, 8-question Mobile Phone Involvement Questionnaire was designed to assess problematic mobile phone use.⁴² Participants who reported having mobile phones completed the Mobile Phone Involvement Questionnaire in Year 2 of follow-up. To quantify the extent of problematic mobile phone use, a sum score was calculated, which was then divided by the number of questions to compute an average score for each participant (range 1–7). Cronbach's alpha was 0.84, indicating good internal consistency.

The self-report, 6-question Social Media Addiction Questionnaire was used to assess problematic social media use. The questions were modeled after the Bergen Facebook Addiction Scale.⁴³ Participants who reported having at least 1 social media account completed the Social Media Addiction Questionnaire in Year 2 of follow-up. To quantify the extent of problematic social media use, a sum score was calculated, which was then divided by the number of questions to compute an average score for each participant (range 1–6). Cronbach's alpha was 0.80, indicating good internal consistency.

The self-report 6-question Video Game Addiction Questionnaire was used to assess problematic video

game use. The questions were modeled after the Bergen Facebook Addiction Scale.⁴³ Participants who reported any video game use completed the Video Game Addiction Questionnaire in Year 2 of follow-up. To quantify the extent of problematic video game use, a sum score was calculated, which was then divided by the number of questions to compute an average score for each participant (range 1–6). Cronbach's alpha was 0.80, indicating good internal consistency.

Regarding mental health outcomes, the Child Behavior Checklist (CBCL) is a screening tool consisting of 112 items asking a parent/caretaker about multiple psychiatric symptoms and behavior problems in their child.^{44,45} The CBCL includes 6 DSM-Oriented Scales: depressive, anxiety, somatic, attention/deficit, oppositional defiant, and conduct problems. During Years 2 and 3 of follow-up, participants' parents/caregivers responded to statements about their child's behavior using a scale from 0 (not true) to 2 (very true/often true) over the past 6 months. T-scores were calculated on the basis of the CBCL scoring rubric.⁴⁶ For this analysis, the CBCL t-scores as a continuous outcome was used, with higher scores indicating a greater number of psychiatric symptoms and behavior problems in children.

The Kiddie Schedule for Affective Disorders and Schizophrenia, a widely used diagnostic tool for assessing symptoms of child and adolescent mental health disorders in the DSM-5, was used for the assessment of suicidal behaviors.^{11,47} The Kiddie Schedule for Affective Disorders and Schizophrenia diagnostic tool for suicide behaviors was given to participants during Years 2 and 3 of follow-up. Participants who reported passive suicidal ideation, nonspecific active suicidal ideation, active suicidal ideation with a plan/method/preparation/intent, or suicide attempt were coded as having suicidal behaviors.

During Years 2 and 3 of follow-up, a 26-item Sleep Disturbance Scale for Children measure was administered to parents/caregivers to assess whether participants had any sleep disorders and measure their overall sleep disturbance, which was summarized into a continuous total score. A cutoff of 39 was used to indicate that a child had overall sleep disturbance, as recommended.⁴⁸ Participants were categorized as having or not having sleep disturbance.

Participants completed the Munich Chronotype Questionnaire to assess their sleep behaviors during follow-up Years 2 and 3.⁴⁹ For the current analyses, the weighted average sleep duration (weekdays and weekends) was used, which was calculated using the sleep durations from both the free days and school days recorded in the Munich Chronotype Questionnaire.

Regarding substance use initiation, participants completed the ABCD Youth Substance Use Introduction

and Patterns survey during follow-up Years 2 and 3 to assess patterns of use for alcohol, tobacco products, and marijuana.⁵⁰ Participants' initiation of substance use was defined as having tried a sip of alcohol, a puff of tobacco product, or a puff of marijuana (present/absent for each substance).⁵¹ In addition, a summary substance initiation measure for any or a combination of substances was included (any substance initiation present versus any substance initiation absent) as an outcome at Year 3.

Regarding covariates, sex assigned at birth (female or male), age of participant, race and ethnicity (Asian, Black, Latino or Hispanic, Native American, other, and White), household income, and parental education status were self-reported by the participants and/or their caregivers. All covariates were collected at the Year 2 assessment. Study site was included to account for regional variation.

Statistical Analysis

Data analyses were performed using STATA 18 (StataCorp, College Station, TX). Descriptive statistics were calculated by measuring the mean, SD, and percentages of each variable. Multiple logistic regression models were conducted to estimate the associations between the exposures (problematic mobile phone, social media, and video game use at Year 2) and the binary outcomes (suicidal behaviors, total sleep disturbance, alcohol initiation, tobacco initiation, marijuana initiation, and any substance use initiation at Year 3), adjusting for covariates and the respective outcome at Year 2. Multiple linear regression models were conducted to estimate the association between the aforementioned exposures and the continuous outcomes (CBCL measures and nightly sleep duration), adjusting for covariates and the respective outcome at Year 2. Assumptions of logistic and linear regression were checked. Sampling weights based on the American Community Survey of the U.S. Census were applied.⁵² To account for multiple statistical testing, the Benjamini–Hochberg procedure was used to adjust for a false discovery rate.⁵³

RESULTS

The analytic sample included 8,119 participants with sociodemographic, predictor, and outcome data. Sociodemographic characteristics of the analytic sample and descriptive statistics for outcome measures are shown in Table 1. Nearly half (47.5%) of the participants were female, and 43.8% were non-White. The mean problematic use score was 3.08 (SD=1.10) for mobile phone use, 1.84 (SD=0.89) for social media use, and 2.09 (SD=1.06) for video game use. In Year 3 of assessment, a small percentage of participants reported initiation of alcohol use

Table 1. Sociodemographic Characteristics of ABCD Study Participants Included in the Current Analysis (N=8,119)

Sociodemographic characteristics	Mean (SD)/%
Age at Year 2 (years)	12.04 (0.66)
Sex assigned at birth (%)	
Female	47.5%
Male	52.5%
Race and ethnicity (%)	
Asian	5.1%
Black	14.7%
Latino or Hispanic	19.5%
Native American	3.3%
Other	1.2%
White	56.2%
Household income at Year 2 (%)	
≤\$24,999	13.5%
\$25,000–\$49,999	16.4%
\$50,000–\$74,999	16.9%
\$75,000–\$99,999	15.1%
\$100,000–\$199,999	28.1%
≥\$200,000	10.1%
Parent's highest education at Year 2 (%)	
College education or more	89.5%
High-school education or less	10.5%
Problematic phone use score at Year 2 ^a	3.08 (1.10)
Problematic social media use score at Year 2 ^b	1.84 (0.89)
Problematic video game use score at Year 2 ^c	2.09 (1.06)
CBCL DSM-Oriented Scales at Year 3 (t-scores)	
Depressive problems	54.18 (6.26)
Anxiety problems	53.69 (6.26)
Somatic problems	54.72 (6.18)
Attention/deficit	53.36 (5.38)
Oppositional defiant problems	53.10 (4.99)
Conduct problems	52.34 (4.71)
Kiddie Schedule for Affective Disorders at Year 3 (%)	
Suicide behaviors	2.9%
Sleep behaviors at Year 3	
Average sleep duration (hours per night)	8.91 (1.58)
Total sleep disturbance score	36.30 (8.10)
Total sleep disturbance (%)	25.7%
Substance use initiation at Year 3 (%)	
Alcohol	11.6%
Tobacco	2.5%
Marijuana	1.0%
Total substance use	12.8%

Note: ABCD Study sampling weights were applied on the basis of the American Community Survey from the U.S. Census.

^aAsked among a subset who reported phone use (n=6,158).

^bAsked among a subset who reported social media use (n=4,633).

^cAsked among a subset who reported video game use (n=6,348).

CBCL, Child Behavior Checklist.

(11.6%), tobacco use (2.5%), marijuana use (1.0%), and any substance use (12.8%).

Figures 1 and 2 and Appendix Table 3 (available online) show the results of the linear and logistic regression models examining PSU associations with mental health, sleep, and substance use initiation outcomes, adjusting for covariates. Problematic mobile phone use, social media use, and video game use were all prospectively associated with depressive, attention/deficit, and oppositional defiant problems on the CBCL DSM-oriented scale. Higher problematic mobile phone use, social media use, and video game use were all prospectively associated with higher depressive problems, attention/deficit problems, and oppositional defiant problems. In addition, higher problematic mobile phone and social media use were prospectively associated with greater somatic and conduct problems. Higher problematic phone use, social media use, and video game use were associated with higher odds of developing suicidal behaviors.

Higher problematic use of mobile phones, social media, and video games was all prospectively associated with shorter sleep duration and a higher odds of sleep disturbance. With regard to substance use initiation, higher problematic mobile phone use and social media use were prospectively associated with higher odds of alcohol, tobacco, and marijuana use initiation.

DISCUSSION

In this large, diverse cohort of U.S. early adolescents, PSU was prospectively associated with a wide range of mental and behavioral health outcomes, indicating that early adolescence may be a crucial time for intervention and prevention of the development of poor mental health outcomes, sleep problems, and substance use initiation. Specifically, problematic mobile phone and social media use was prospectively associated with higher odds of depressive, somatic, attention/deficit, oppositional defiant, and conduct problems as well as suicidal behaviors, sleep disturbance, and substance use initiation. Problematic video game use was associated with the outcomes mentioned earlier, except for somatic and conduct problems and substance initiation. Although effect sizes were small relative to SDs, they exceeded those previously reported for screen time and similar mental health outcomes.⁵⁴ A vast majority of participants were under the minimum age of 13 years for social media use,⁵⁵ indicating that social media age verification is not robust.

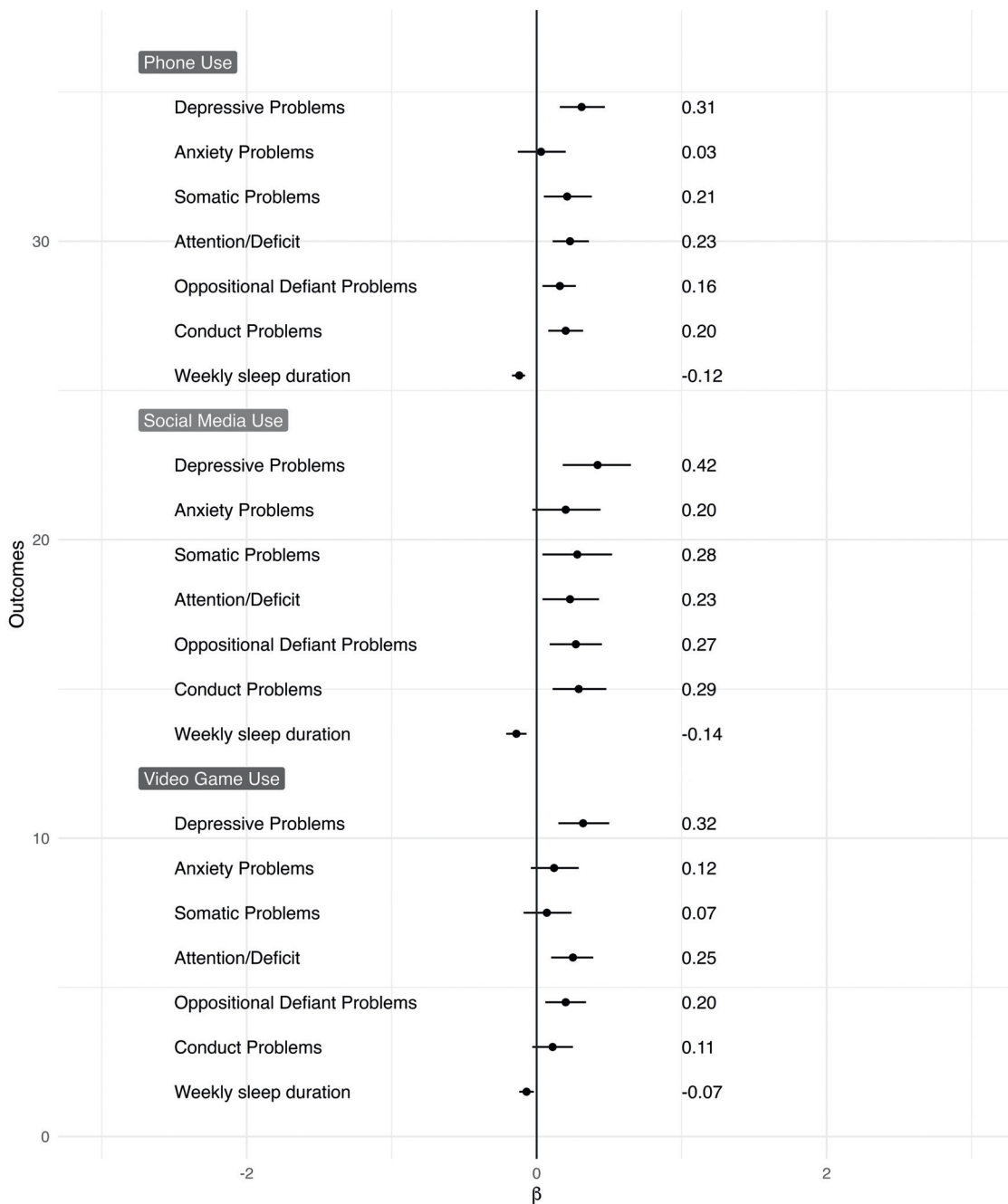


Figure 1. Prospective associations between problematic screen use and continuous mental health and sleep outcomes 1 year later in the ABCD Study (N=8,119).

Note: The figure shows the forest plot of linear regression coefficient and 95% CIs for associations with problematic screen use score (Year 2) and continuous mental health and sleep outcomes (Year 3), adjusted for age, sex, race and ethnicity, household income, parent education, study site, and the respective mental health or sleep measure at Year 2.

ABCD, Adolescent Brain Cognitive Development.

The findings are largely consistent with those of previous studies correlating PSU with depression,^{4,16,19,20} attention-deficit/hyperactivity disorder (ADHD),^{3,16,32,56} aggression and hostility,^{3,32,57,58} and sleep disturbance.^{15,19,20,26} There are a variety of factors that might explain these associations.

Fear of missing out may mediate the relationship between problematic mobile phone and social media use and depression;⁵⁹ video games may be used as a form of escapism,⁶⁰ leading to social isolation and higher risk of depression.⁶¹ Constant notifications and multitasking on mobile phones

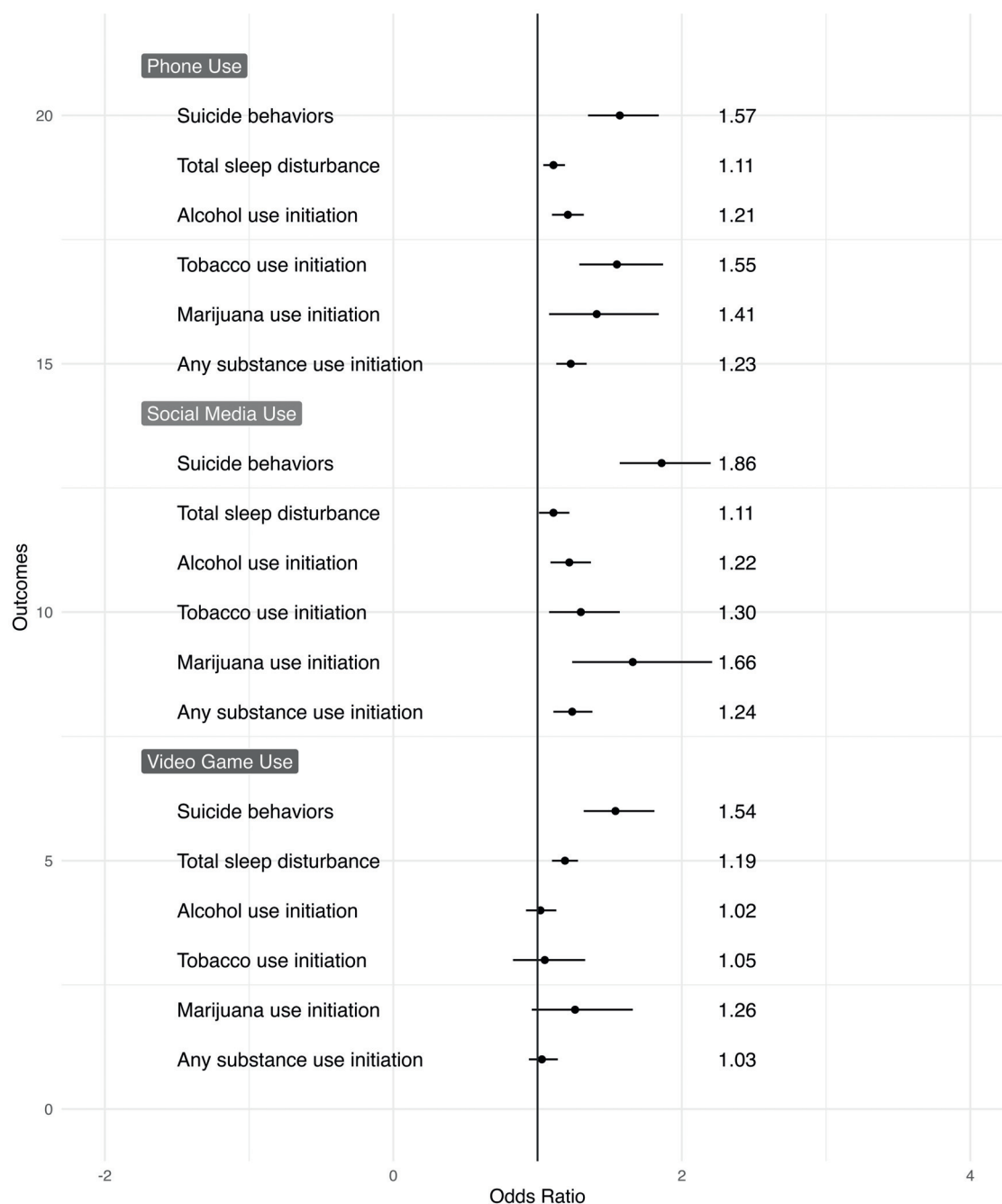


Figure 2. Prospective associations between problematic screen use and binary suicide, sleep, and substance use outcomes 1 year later in the ABCD Study (N=8,119).

Note: The figure shows the forest plot of AORs and 95% CIs from logistic regression models for associations with problematic screen use score (Year 2) and binary suicide, sleep, and substance use initiation outcomes (Year 3), adjusted for age, sex, race and ethnicity, household income, parent education, study site, and the respective suicide, sleep, and substance use measure at Year 2.

ABCD, Adolescent Brain Cognitive Development.

and social media may contribute to their association with ADHD;^{56,62} as for video games, previous research has shown that individuals with ADHD are at higher risk of video game addiction.⁶³ PSU may be linked to aggression through disturbance of sleep quality,^{15,26,57} which has been

shown to affect mood/behavior.^{64,65} Exposure to violent video games may also play a role in normalizing aggression.⁶⁶

Although PSU was associated with higher scores across most mental health measures, the finding that PSU was not

associated with anxiety departs from prior literature.^{16,17,19,24} This may be due to the younger mean age of the study sample, which is less than the average age of onset for anxiety disorders.⁶⁷ There may also be a number of explanations for this nonsignificant finding. Studies have found that social media can help individuals feel more connected and, in turn, reduce stress/anxiety and improve psychological well-being.^{68,69} Anxiety symptoms and social isolation increased among youth during the COVID-19 pandemic,^{70,71} and it is possible—though only speculative—that social media (and by extension, mobile phone use) served as an anxiolytic when baseline anxiety levels were already elevated. In addition, a recent systematic review demonstrated that during the COVID-19 pandemic, playing video games reduced anxiety and loneliness among adolescents.⁷² This study also showed no association between problematic video game use and conduct disorder, whereas previous studies have shown associations with aggression and irritability.²⁵ One prospective study found a weak association between violent video game use (although not necessarily problematic use) at an early age and increased risk of conduct disorder in late adolescence.⁷³ This study examined early adolescents, with the possibility that conduct disorder from problematic use may not arise until later in adolescence. Future research should further examine the associations between PSU and anxiety since the onset of the COVID-19 pandemic as well as prospectively investigate problematic video game use and conduct disorder through late adolescence.

This study also found that problematic mobile phone use and social media use were associated with somatic problems—such as headaches, body aches, and feeling dizzy—which is consistent with the literature.^{74,75–77} In particular, mobile phone and social media use may contribute to somatic symptoms due to neck strain.^{75,78} There was no association between problematic video game use and somatic problems despite previous literature indicating otherwise.^{79,80} With regard to PSU and substance use, this study showed associations between problematic social media use and mobile phone use and substance use initiation, concurring with the literature in the case of social media use¹⁸ and favoring the evidence for an association in the case of mobile phone use, where previous studies have differed.^{5,28} Possible mechanisms for such associations include the significance of peer influence in adolescence as well as increased exposure to substance-related online content.¹⁸ The literature on problematic video game use and substance use is also inconclusive,^{29–31} and in this study, no association was found.

Limitations

The strengths and limitations of this study should be noted. The study adds to knowledge in the field by examining prospective associations between PSU and

mental health and substance use outcomes, whereas a majority of previous studies have been cross-sectional. In addition, strengths include analysis of a large, demographically diverse, contemporary sample of early adolescents in the U.S. Limitations include the observational nature of the study, which precludes causal inferences and leads to susceptibility to residual and unmeasured confounders despite adjustment for potential confounders, as well as reporting, recall, and social desirability bias.⁸¹ In addition, the study relies on self/parent-report rather than objective measures of use or diagnoses and uses mean scores instead of thresholds for PSU because there are no validated thresholds for the measures used. The study also includes a highly educated sample and has a relatively short 1-year follow-up period. It is worth noting that the study showed small effect sizes, especially with regard to SDs and relative to the change required in problematic screen behaviors.

CONCLUSIONS

This study's findings have potential implications for health policy and clinical practice. The findings highlight the possible benefit of interventions specifically tailored to early adolescents because this is a critical period where mental health concerns begin to develop.⁸ Clinicians can consider inquiring about PSU and screen time in early adolescents, providing anticipatory guidance, media literacy education, or referrals to therapy as needed.⁸² Importantly, problematic use is a modifiable exposure, amenable to both behavioral interventions and changes in digital technology design. Professional organizations such as the American Academy of Pediatrics can consider refining guidelines regarding screen time and continue to encourage families to utilize a family media use plan.⁸³ Future research should continue to examine the prospective relationships between and potential bidirectionality of PSU and mental health and substance use outcomes as the cohort ages to mid and late adolescence as well as aim to elucidate the mechanisms underlying the associations.

ACKNOWLEDGMENTS

The authors thank Christiane Helmer and Anthony Kung for analytic and editorial assistance.

Funding: JMN was supported by the NIH (K08HL159350). FCB was supported by the NIH (R01DA057567).

Declaration of interest: None.

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SUPPLEMENTAL MATERIAL

Supplemental materials associated with this article can be found in the online version at <https://doi.org/10.1016/j.amepre.2025.108248>.

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